Complexified Cube Chains and a Braid Invariant of Precubical Sets

July 31, 2026

Abstract

Ziemiański's cube chain category $\mathrm{Ch}(K)$ is a combinatorial model for the space of directed paths on a precubical set K, and Paliga and Ziemiański build a braid invariant out of $\pi_1|\mathrm{Ch}(Z)|$. That invariant trivializes for Euclidean K because $|\mathrm{Ch}(I^n)|$ is contractible [PZ21, Prop. 6.1]. Following an analogy from the hyperplane arrangement literature, we build a "complexified" version of Ch, which we call Ch^*. It does not collapse: in fact $\mathrm{Ch}^*(I^n) \cong \mathrm{Sal}(n)^{\mathrm{op}}$, the Salvetti poset of the braid arrangement. Salvetti's theorem tells us $|\mathrm{Ch}^*(I^n)| \simeq \mathrm{Conf}(n, \mathbb{C})$, which is not contractible. In fact, its fundamental group is the pure braid group P_n. The construction of Ch^* works by attaching one dimensional refinements to each cube chain in K. Then morphisms induce permutations, which give us a functor $\mathrm{Conc} : \mathrm{Ch}^*(K) \to \mathbf{Braid}^+$ into the positive braid monoid. It is natural in K, and non-trivial as long as K contains a square. The construction and the comparison with $\mathrm{Sal}(n)$ are formalized in Lean [aut26].

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1 Introduction

1.1 The problem

Ziemiański's cube chain category $\mathrm{Ch}(K)$ is a combinatorial model for the executions of a bipointed precubical set K [Zie20, Thm. 7.6], and Paliga and Ziemiański use it to build braid group invariants of K — representations $IB_n(K,-)$ into B_n, and characteristic classes — by transporting along the unique map $K \to Z$ into the final precubical set [PZ21, §6]. On the class most concurrency examples live in, these vanish.

Theorem 1.1 ([PZ21, Prop. 6.1])**.** *Let $K \subseteq I^n$ — a* sculpture *[Pra91], equivalently a* Euclidean cubical complex *[RZ14]. Then $IB_n(K,-)$ and all the associated characteristic classes are trivial.*

The proof is one line: the invariant factors through $|\mathrm{Ch}(I^n)|$, which is contractible.

1.2 The proposal

The same phenomenon is familiar for real hyperplane arrangements: the order complex of the face poset of a real arrangement is contractible, and the fix is to complexify. The Salvetti poset [Sal87; Par93; BZ92] is the solution, a complex whose cells are a face together with a chamber above it.

We carry that fix over. The analogy will turn into an equivalence for cubes, but the construction will work for any precubical set. Chambers, also known as topes, in hyperplanes are faces that cannot be refined further. These correspond to one dimensional cube chains, we call *runs*, in a precubical set. So we decorate each cube chain with a run refining it, exactly as Salvetti decorates each face with a tope above it. Definition 5.8 does this for an arbitrary precubical set, by exhibiting runs as a presheaf and taking a category of elements.

1.3 Results

Theorem 1.2 (Theorem 5.11)**.** *For every $n \geq 0$ there is an isomorphism of posets $\mathrm{Ch}^*(I^n) \cong \mathrm{Sal}(n)^{\mathrm{op}}$, where $\mathrm{Sal}(n)$ is the Salvetti poset of the braid arrangement A_{n-1}. Consequently $|\mathrm{Ch}^*(I^n)| \simeq \mathrm{Conf}(n,\mathbb{C})$ and $\pi_1|\mathrm{Ch}^*(I^n)| \cong P_n$, the pure braid group on n strands.*

So indeed this resolves the issue where contractability of $\mathrm{Ch}(I^n)$ kills the invariants. The braid data can also be extracted directly, without passing through Salvetti's theorem at all.

Theorem 1.3 (Theorem 6.8, Proposition 6.10)**.** *For every bipointed precubical set K there is a functor $\mathrm{Conc} : \mathrm{Ch}^*(K) \to \mathbf{Braid}^+$ into the positive braid monoids, sending a morphism to the germ of the permutation by which it reorders events. It is natural in K, and it is not constant at the identity braid as soon as there is a map $I^2 \to K$.*

This rests on a "no double crossing" property (Lemma 6.6): along a composite in $\mathrm{Ch}^*(K)$ no two events cross twice, so the associated permutations are length-additive and multiply in the Garside germ presentation [Gar69; Deh+15]. Unlike IB_n, the result is basepoint-free and lands over the ordered rather than the unordered configuration space; see Remarks 6.11 and 6.12.

1.4 Formalisation

Everything above except the topology is formalised in Lean 4 on top of mathlib [aut26]: the presheaf ρ, the category $\mathrm{Ch}^*(K)$, its projection to $\mathrm{Ch}(K)$, the comparison of Theorem 1.2, and the braid functor of Theorem 1.3.

1.5 Notation

To fix notation, let **Cube** be the box category: its objects are the cubes I^n for $n \geq 0$, and its morphisms are the face embeddings (composites of the coface maps δ_i^ϵ). A *precubical set* is a presheaf on **Cube**; write $\square$ for the category of precubical sets. Concretely, a precubical set K has, for each $n \geq 0$, a set K_n of n-cells, together with face maps

$$d_i^\epsilon : K_n \to K_{n-1} \qquad (1 \leq i \leq n,\ \epsilon \in \{0,1\}),$$

where i names a direction and ϵ a side, subject to the precubical identities $d_i^\epsilon d_j^\eta = d_{j-1}^\eta d_i^\epsilon$ for $i < j$. We write I^n both for the standard n-cube as an object of **Cube** and for the presheaf it represents, and we set $[n] := \{1, \dots, n\}$, so that $[0] = \emptyset$. Yoneda identifies the two descriptions of a cell,

$$K_n \ = \ K(I^n) \ = \ \mathrm{Hom}_\square(I^n, K),$$

and we pass between them without comment; in particular K_0 is the set of vertices.

We usually work with precubical sets carrying designated start and end vertices. A *bipointed* precubical set is a precubical set K with two chosen vertices $\mathrm{init}_K, \mathrm{fin}_K \in K_0$, and morphisms are required to preserve them; write $\square_*$ for the resulting category. Each cube I^n is bipointed by its initial vertex $(0, \dots, 0)$ and final vertex $(1, \dots, 1)$. We write Z for the final object of $\square_*$.

Convention 1.4. Named ambient categories are set in bold (**Cat**, **Set**, **Cube**, **Braid**); constructions applied to an object are set upright ($\mathrm{Ch}(K)$, $\mathrm{Run}(K)$, $\mathrm{Sal}(n)$). Functors are written with $\to$; we reserve $\Rightarrow$ for natural transformations. The standard n-cube is I^n; the symbol $\square$ names the category of precubical sets.

Remark 1.5 (Variance)*.* The $^{\mathrm{op}}$ in Theorem 1.2 is merely an artefact of choices made by different literature we reference. That is, in the cube chain literature [PZ21; Zie20] a morphism of $\mathrm{Ch}(K)$ runs from a *finer* chain to a *coarser* one (Definition 2.1). And in the arrangement literature [Sal87; Par93] the face order runs from *smaller* faces to *larger* ones.

2 Background: cube chains

We recall three constructions on $\square_*$ from [PZ21; Zie20].

The first is the **wedge** $\vee$. Given $P, Q \in \square_*$, the wedge $P \vee Q$ is the pushout, formed in $\square$, gluing P's final vertex to Q's initial vertex:

$$\begin{tikzcd} I^0 \arrow[r, "\mathrm{init}_Q"] \arrow[d, "\mathrm{fin}_P"'] & Q \arrow[d] \\ P \arrow[r] & P \vee Q \arrow[ul, phantom, "\ulcorner", very near start] \end{tikzcd}$$

Note that the two legs are maps of $\square$, not of $\square_*$: a bipointed map $I^0 \to Q$ would have to send the point to both init_Q and fin_Q. The wedge $P \vee Q$ is bipointed by init_P and fin_Q, and $(\square_*, \vee, I^0)$ is monoidal with unit the point I^0. The **serial wedge** of a list of *positive* naturals is

$$\bigvee [a_1, \dots, a_r] := I^{a_1} \vee \cdots \vee I^{a_r},$$

with $\bigvee[\,] = I^0$. Write $\square^\vee \subseteq \square_*$ for the full subcategory of serial wedges.

Definition 2.1 ([PZ21, §2]). The second construction: For $K \in \square_*$, let $\mathrm{Ch}(K)$ be the full subcategory of the slice $\square_*/K$ spanned by the serial wedges: an object is a bipointed map $\bigvee a \to K$, and a morphism $(\bigvee a \to K) \to (\bigvee b \to K)$ is a bipointed map $\bigvee a \to \bigvee b$ commuting over K. Such a morphism runs from a finer chain to a coarser one.

The third construction is **refinement**. A *chain* in K is a finite sequence $[c_1, \ldots, c_l]$ of cells of K, each of dimension at least 1, such that $\mathrm{fin}(c_i) = \mathrm{init}(c_{i+1})$ for all i, with $\mathrm{init}(c_1) = \mathrm{init}_K$ and $\mathrm{fin}(c_l) = \mathrm{fin}_K$; here $\mathrm{init}(c)$ and $\mathrm{fin}(c)$ denote the images under $c : I^{\dim c} \to K$ of the initial and final vertices of the cube. A *refinement* replaces a cell c_i by a chain of the cube $I^{\dim c_i}$, post-composed with c_i. Let $\mathrm{Refine}(K)$ be the category of chains, with morphisms generated by refinements and oriented as in $\mathrm{Ch}(K)$ (finer to coarser).

Finally, two properties of K from [PZ21] recur. We say K *admits altitude* if there is a function $\mathrm{alt} : K_0 \to \mathbb{Z}$ with $\mathrm{alt}(\mathrm{fin}(e)) = \mathrm{alt}(\mathrm{init}(e)) + 1$ for every edge $e \in K_1$. Following [PZ21, §2] verbatim, K is *non-self-linked* (NSL) if every $\square$-map $I^n \to K$ is injective, for all $n \geq 0$. By Yoneda a $\square$-map $I^n \to K$ is an n-cell of K, so this says the iterated faces of every cell are pairwise distinct.

We use the following, all from [PZ21].

Lemma 2.2 ([PZ21, §2]). *The $\vee$ operator is a monoidal product on $\square_*$, with unit I^0.*

Lemma 2.3. $\mathrm{Ch} : \square_* \to \mathbf{Cat}$ *is a lax monoidal functor from $(\square_*, \vee)$ to $(\mathbf{Cat}, \times)$: the comparison $\mathrm{Ch}(P) \times \mathrm{Ch}(Q) \to \mathrm{Ch}(P \vee Q)$ concatenates chains.*

Lemma 2.4 ([PZ21, Lem. 2.11]). *Let K admit altitude and be NSL. Then $\mathrm{Ch}(K)$ is a poset, every chain is determined by its set of cubes, and every bipointed map $\bigvee a \to K$ is injective. Consequently $\mathrm{Ch}(K)$ and $\mathrm{Refine}(K)$ are equivalent.*

Futhermore, a the comparison of Lemma 2.3 is an isomorphisms of categories, since both categories are skeletal and posets.

Lemma 2.5. *Serial wedges admit altitude and are NSL. Explicitly,* alt *counts edges from* init*, so the final vertex of $\bigvee a$ has altitude $\sum_i a_i$, and the vertices of the j-th summand are exactly those of altitude in $[s_j^a,\ s_{j+1}^a]$, where $s_j^a := \sum_{j'<j} a_{j'}$.*

Lemma 2.6 ([PZ21, Prop. 2.7]). *Let P, Q be connected bipointed precubical sets admitting altitude functions. Then every bipointed map $f : P \to Q$ preserves altitude:* $\mathrm{alt}_Q \circ f = \mathrm{alt}_P$.

Proof sketch. A precubical map takes edges to edges, so alt_Q increases by exactly one along the image of each edge; connectivity propagates this, and $f(\mathrm{init}_P) = \mathrm{init}_Q$ pins the constant. □

This is all the foundational material we need.

3 Wedge machinery

Most of the technical material in this paper involves bookkeeping around coordinates of wedges. So we build a bit of machinery here to assist.

Lemma 3.1. *A functor $P : \mathbf{Cube} \to \mathbf{Set}$ extends, by left Kan extension along the Yoneda embedding, to a cocontinuous $E(P) : \square \to \mathbf{Set}$ with $E(P)(I^n) = P(I^n)$. If $P(I^0) = \emptyset$ then for every serial wedge*

$$E(P)\Big(\bigvee a\Big) \;\cong\; \coprod_i P(I^{a_i}). \tag{1}$$

Proof. Applying $E(P)$ to the pushout defining $A \vee B$ gives $E(P)(A \vee B) = E(P)(A) \sqcup_{P(I^0)} E(P)(B)$, which is the coproduct when $P(I^0) = \emptyset$. $\square$

Lemma 3.2. *Let P be a precubical set and put $E(P) := \mathrm{Hom}_\square(-, P) : \square^{\mathrm{op}} \to \mathbf{Set}$, the right Kan extension of P along Yoneda. If P_0 is a singleton then for every serial wedge*

$$E(P)\Big(\bigvee a\Big) \;\cong\; \prod_i P_{a_i}. \tag{2}$$

Proof. A hom functor takes colimits in its first variable to limits, so $E(P)$ sends the pushout defining $A \vee B$ to the pullback $E(P)(A) \times_{E(P)(I^0)} E(P)(B)$. Here $E(P)(I^0) = P_0$ is a singleton, hence terminal in **Set**, and a pullback over a terminal object is a product. $\square$

Definition 3.3. Let $\mathrm{Coord} : \mathbf{Cube} \to \mathbf{Set}$ send the cube I^n to its set of directions $[n]$, and send a face embedding $g : I^n \to I^m$ to the monotone injection $g_* : [n] \hookrightarrow [m]$ that names the n directions g leaves free. Since $\mathrm{Coord}(I^0) = [0] = \emptyset$, Lemma 3.1 applies:

$$E(\mathrm{Coord})\Big(\bigvee a\Big) \;\cong\; \coprod_i [a_i].$$

For a morphism $f : \bigvee a \to \bigvee b$ we write $f_* := E(\mathrm{Coord})(f)$.

Definition 3.4. Write $\mathrm{blk}(c)$ for the summand index of a coordinate $c \in \coprod_i [a_i]$. By the pushout defining $\bigvee a$, a morphism $f : \bigvee a \to \bigvee b$ is determined by the $a_i \to \bigvee b$. But each such image lies in a single I^{b_j} since $a_i > 0$. So it corresponds to a order preserving face embedding

$$f_i : I^{a_i} \longrightarrow I^{b_{\mathrm{blk}(f)(i)}}.$$

This defines $\mathrm{blk}(f)$, and describes f_* on coordinates:

$$f_*(i, d) \;=\; \big(\mathrm{blk}(f)(i),\; (f_i)_*(d)\big), \qquad d \in [a_i]. \tag{3}$$

In particular $\mathrm{blk} \circ f_* = \mathrm{blk}(f) \circ \mathrm{blk}$.

Lemma 3.5. *For any morphism $f : \bigvee a \to I^m$ in $\square$, the function f_* is injective.*

Proof. Induct on a; the empty case is $f_* : [0] \to [m]$. For the step, write $f : (\bigvee a) \vee I^n \to I^m$, which by the pushout splits into $g : \bigvee a \to I^m$ and $h : I^n \to I^m$ with $f_* = g_* \sqcup h_*$ by (Lemma 3.1). Here g_* is injective by hypothesis, and h_* is the monotone injection attached to the face embedding h.

So it suffices that $\mathrm{im}\, g_*$ and $\mathrm{im}\, h_*$ are disjoint. Let $c \in \{0,1\}^m$ be the image of the glued vertex $\mathrm{fin}(\bigvee a) = \mathrm{init}(I^n)$. Coordinates only increase along a map into a cube, so every direction g leaves free is already 1 at c, while every direction h leaves free is still 0 at c. $\square$

Lemma 3.6. *For any morphism $f : \bigvee a \to \bigvee b$ in $\square_*$, the function f_* is bijective and $\mathrm{blk}(f)$ is monotone.*

Proof. To prove injective and monotone, we go by induction on the length of b, generalising over a. If b is empty, then so is a by Lemma 2.6 and Lemma 2.5. So both $\mathrm{blk}(f)$ and f_* are the empty function.

Now consider $f : \bigvee a \to I^m \vee \bigvee b$. By Lemma 2.5 and Lemma 2.4, $\mathrm{Ch}(I^m) \times \mathrm{Ch}(\bigvee b) = \mathrm{Ch}(I^m \vee \bigvee b)$. We can view f as an object of $\mathrm{Ch}(I^m \vee \bigvee b)$. So a splits as $a = a_l * a_r$, and f splits as

$$(f_1 : \bigvee a_l \to I^m, f_2 : \bigvee a_r \to \bigvee b)$$

Since $f = f_1 \vee f_2$ and $E(\mathrm{Coord})$ is cocontinuous (Lemma 3.1),

$$f_* = (f_1 \vee f_2)_* = (f_1)_* \sqcup (f_2)_*$$

Now $(f_2)_*$ is injective by the induction hypothesis and $(f_1)_*$ is injective by Lemma 3.5. This is a disjoint union of injections with disjoint codomains, hence f_* is injective. For blk, note

$$\mathrm{blk}(f) = \mathrm{blk}(f_1) \sqcup (1 + \mathrm{blk}(f_2))$$

Now $\mathrm{blk}(f_1)$ is always 1, and $1 + \mathrm{blk}(f_2)$ is monotone by induction, and > 1. So $\mathrm{blk}(f)$ is monotone as well.

For surjectivity of f_*, we count. By Lemma 2.5 the final vertex of $\bigvee a$ has altitude $\sum_i a_i$, and likewise for b; by Lemma 2.6 the bipointed map f preserves altitude so $\sum_i a_i = \sum_j b_j$. So f_* is an injective function between finite sets with equal cardinality. So f_* is a bijection. □

4 The braid arrangement and the Salvetti poset

The construction of Ch^* ports the following technique to precubical sets.[1] We describe the technique here, following Paris [Par93], and build Ch^* in the next section.

Let $\mathrm{Face}(n)$ be the set of total preorders on $[n]$. A total preorder F, induces an equivalence relation $\sim_F$, which are linearly ordered by $<_F$. The classes of $\sim_F$ are called blocks, which give an ordered set partition of $[n]$. Order $\mathrm{Face}(n)$ by

$$F \leq G \quad := \quad <_F \; \subseteq \; <_G,$$

This is the poset of **covectors**, or faces, of the braid arrangement $A_{n-1} = \{ x_i = x_j \}_{i<j}$ in $\mathbb{R}^n$. A point x gives the covector $i \leq_F j := x_i \leq x_j$, the weak order of its coordinates. The maximal covectors are those with no ties, that is the linear orders on $[n]$; these are the **topes**, and they correspond to the components of $\mathbb{R}^n \setminus \bigcup A_{n-1}$.

Definition 4.1. The **Salvetti poset** is

$$\mathrm{Sal}(n) := \{ (F, C) \;\mid\; F \in \mathrm{Face}(n),\ C \text{ a tope},\ F \leq C \},$$

We define a composition of faces as $F \circ G$ by $i \leq_{F \circ G} j$ iff $i <_F j$, or $i \sim_F j$ and $i \leq_G j$. In particular $F \leq F \circ G$, and $F \circ C$ is again a tope whenever C is. We give $\mathrm{Sal}(n)$ a partial order structure

$$(F, C) \preceq (F', C') \quad :\Longleftrightarrow \quad F \leq F' \text{ and } C' = F' \circ C.$$

This is the order of [Par93]; see [DDP25] for the form used in the formalisation.

Theorem 4.2 (Salvetti [Sal87]; see also [BZ92; Par93])**.** *$|\mathrm{Sal}(n)| \simeq \mathrm{Conf}(n, \mathbb{C}) = \{z \in \mathbb{C}^n : z_i \neq z_j$ for $i \neq j\}$. This is ordered the configuration space on n points, as well as the complement of the complexified A_{n-1}. So $\pi_1|\mathrm{Sal}(n)| \cong P_n$, the pure braid group.*

[1] The lean repo uses Conditional Oriented Matriods instead of ordered set partitions to define the braid arrangement, but proves they are equivalent

The packaging we will imitate is as a Grothendieck construction. Let

$$\mathrm{Tope} : \mathrm{Face}(n) \to \mathbf{Set}, \qquad \mathrm{Tope}(F) := \{C \text{ a tope} : F \le C\},$$

be the covariant functor sending $F \le F'$ to the map $C \mapsto F' \circ C$. Then

$$\mathrm{Sal}(n) \;\cong\; \textstyle\int \mathrm{Tope},$$

the Grothendieck construction: its objects are the pairs (F, C) with $C \in \mathrm{Tope}(F)$, and its morphisms are exactly the relations $\preceq$ above. Intuitively, each face is annotated with a compatible tope, a local orientation; crossing a face can flip the orientation, and these flips are the braid letters.

5 The complexification Ch^*

The first observation that suggests a bridge is feasible here is that

Lemma 5.1. $\mathrm{Ch}(I^n)^{\mathrm{op}} \cong \mathrm{Face}(n)$

By Lemma 3.6 a chain of I^n is just an ordered set partition of its directions, and a morphism of chains is a refinement. The rest is rote, and formalized in Lean. But, to establish an analogy that works for any precubical set, we need to produce a precubical notion of tope.

5.1 Runs

Definition 5.2. A serial wedge is **one dimensional** if all its entries are 1.

Definition 5.3. Let $\mathrm{Run}(K)$ be the set of one dimensional chains in $\mathrm{Ch}(K)$.

Lemma 5.4. Run *Forms a monoidal functor from lists of positive integers to Set*

Proof. Note that wedge respects one dimensionality, so $\mathrm{Run}(K)$ is can also be seen as a full, monoidal subcategory of $\mathrm{Ch}(K)$. So Run itself becomes a monoidal functor. □

Now, we want a contravariant operation from $\mathrm{Ch}(K)$ to runs, mirroring the order from Salvetti:

$$(\bigvee a \to \bigvee b) \;\longmapsto\; (\mathrm{Run}(\bigvee b) \to \mathrm{Run}(\bigvee a)).$$

We will define a functor $\rho : \mathbf{Cube}^{\mathrm{op}} \to \mathbf{Set}$ which will give us this shape. First, note that every morphism $f : I^n \to I^m$ is a face embedding. So has a corresponding projection on cells $\hat{f} : I^m \to I^n$. Notably, the projection is monotolically decreasing on cell dimension. With that, we can define ρ as follows

Definition 5.5. On objects, let $\rho(I^n) = \mathrm{Run}(I^n)$. By the pushout defining the wedge, a run of I^m is a list $[c_1, \ldots, c_m]$ of composable edges from init to fin. For a face embedding $f : I^n \to I^m$, let

$$\rho(f)[c_1, \ldots, c_m] \;:=\; [\,\hat{f}(c_1), \ldots, \hat{f}(c_m)\,] \quad \text{with the 0-cells deleted.}$$

This is again a run: $\hat{f}(c_k)$ will preserve exactly n coordinates, giving n cells. The projection preseves boundary maps, so even after deleting 0 cells, it remains a wedge map. And $\hat{f}$ carries init to init and fin to fin.

In coordinates, $r_*(k)$ is the free direction of c_k, so the surviving steps are those in $S := r_*^{-1}(\mathrm{im}\, f_*)$, and they keep their relative order. Writing $\iota : [n] \to S$ for the increasing bijection,

$$\rho(f)(r)_* \;=\; f_*^{-1} \circ r_* \circ \iota. \tag{4}$$

Lemma 5.6. *$\rho : \mathbf{Cube}^{\mathrm{op}} \to \mathbf{Set}$ is a functor and $\rho(I^0)$ is a singleton; so ρ is a precubical set with a single vertex.*

Proof. The projection of the identity embedding is the identity. And projections compose to projections, so ρ is a functor. The empty wedge map is the sole member of $\rho(I^0)$. □

So ρ is a precubical set, and we write $E(\rho) := \mathrm{Hom}_{\square}(-, \rho) : \square_*^{\mathrm{op}} \to \mathbf{Set}$.

Lemma 5.7 (ρ classifies runs)**.** *For every serial wedge $\bigvee a$ there are bijections*

$$E(\rho)(\textstyle\bigvee a) \;=\; \mathrm{Hom}_{\square}(\bigvee a,\; \rho) \;\cong\; \prod_i \mathrm{Run}(I^{a_i}) \;\cong\; \mathrm{Run}(\bigvee a).$$

Proof. The left congruence is Lemma 3.2, since ρ_0 is a singleton (Lemma 5.6). The right is the monoidal property of Run. □

We will freely identify $E(\rho)(\bigvee a)$ with $\mathrm{Run}(\bigvee a)$.

5.2 The construction

Definition 5.8. Let $W(K) : \mathrm{Ch}(K) \to \square^{\vee}$ be the forgetful functor $(\bigvee a \to K) \mapsto \bigvee a$. Set

$$\mathrm{Lines}(K) := E(\rho) \circ W(K)^{\mathrm{op}},$$

a presheaf on $\mathrm{Ch}(K)$, and

$$\mathrm{Ch}^*(K) := \textstyle\int \mathrm{Lines}(K),$$

the category of elements of $\mathrm{Lines}(K)$, with projection $\pi : \mathrm{Ch}^*(K) \to \mathrm{Ch}(K)$.

Defining Ch^* abstractly has some conveniences later. Let us unpack the definition of Ch^* to have a more concrete presentation of it.

Objects of $\mathrm{Ch}^*(K)$ are pairs

$$(\bigvee a \to K,\; x \in \mathrm{Run}(\bigvee a)), \;\text{ i.e. }\; \bigvee[1, \ldots, 1] \to \bigvee a \to K,$$

a cube chain together with a linear ordering of its events. The intuition here matches the Salvetti story precisely: we have a wedge map into K, coupled with orientation info about it. Morphisms are a little uglier. A morphism

$$(\bigvee a \to K,\; x \in \mathrm{Run}(\bigvee a)) \longrightarrow (\bigvee b \to K,\; y \in \mathrm{Run}(\bigvee b))$$

unpacks to

1. A morphism $f : (\bigvee a \to K) \to (\bigvee b \to K)$, i.e. a bipointed map $f' : \bigvee a \to \bigvee b$ commuting over K.
2. The condition $x = \mathrm{Lines}(f)(y)$: we take the run y, project it backwards along $\hat{f}$, and require the result to be x.

The diagram below records the story. The solid arrows are morphisms of $\mathrm{Ch}^*(K)$; the dashed lines are not. $E(\rho)(f)$ is the projection associated with f. And $\sigma(f)$ is an induced permutation we will study in the next section

$$\begin{tikzcd}
\bigvee[1,\dots,1] \arrow[r, "x"] \arrow[d, dashed, "\sigma(f)"'] & \bigvee a \arrow[d, "f"] \arrow[d, dashed, bend right=60, "E(\rho)(f)"'] \arrow[dr] & \\
\bigvee[1,\dots,1] \arrow[r, "y"'] & \bigvee b \arrow[r] & K
\end{tikzcd}$$

5.3 The crossing permutation

Let's look more closely at how σ is constructed. Let $N = \Sigma_i a_i = \Sigma_j b_j$. Then define

$$\sigma(f) := y_*^{-1} \circ f'_* \circ x_*$$

and observe that by Lemma 3.6, $\sigma(f)$ is a composition of bijections, so $\sigma(f) \in \Sigma_n$. Also, the form $y_* \circ \sigma(f) = f'_* \circ x_*$ will be convenient. Lemma 3.6 the source and target of f have the same number of coordinates. This permutation $\sigma(f)$ is what will power our braid functor.

Lemma 5.9. *$\sigma : \mathrm{Ch}^*(K) \to \bigsqcup_N \Sigma_N$ is a functor, where the target is the groupoid with objects the naturals and $\mathrm{Hom}(N, N) = \Sigma_N$.*

Proof. Along a composite the middle ordering y_* cancels,

$$\sigma(g \circ f) = z_*^{-1} \circ g'_* \circ f'_* \circ x_* = (z_*^{-1} \circ g'_* \circ y_*)(y_*^{-1} \circ f'_* \circ x_*) = \sigma(g)\,\sigma(f),$$

and

$$\sigma(\mathrm{id}) = x_*^{-1} \circ (\mathrm{id})_* \circ x_* = \mathrm{id}\,. \qquad \square$$

Lemma 5.10. *Let $f : (\bigvee a, x) \to (\bigvee b, y)$ in $\mathrm{Ch}^*(K)$ and suppose $\mathrm{blk}(x_*(k)) = \mathrm{blk}(x_*(l))$. Then $k < l$ if and only if $\sigma(f)(k) < \sigma(f)(l)$.*

Proof. Let $i = \mathrm{blk}(x_*(k))$. Then $x \in \mathit{Run}(\bigvee a) \simeq \prod_j \mathrm{Run}(I^{a_j})$, by Lemma 5.4. So let x_i be its i^{th} component. Likewise, let $j = \mathrm{blk}(f')(i)$, and y_j be its j^{th} component.

Now, since x is one dimensional, $\mathrm{blk} \circ x_* = \mathrm{blk}(x)$, which is monotone (Lemma 3.6). So $x_*^{-1}(i)$ is an interval, order-isomorphic to the positions $[a_i]$ of x_i, and likewise for y and $[b_j]$.

The morphism condition restricts to $x_i = \rho(f_i)(y_j)$ (Definition 3.4), so $\sigma(f) = y_*^{-1} \circ f'_* \circ x_*$ restricts to $(y_j)_*^{-1} \circ (f_i)_* \circ (x_i)_*$, which by (4) is the increasing inclusion $[a_i] \to [b_j]$. Both identifications preserve order, so $k < l$ if and only if $\sigma(f)(k) < \sigma(f)(l)$. $\square$

5.4 Comparison with the Salvetti poset

Theorem 5.11. *For every $n \geq 0$ there is an isomorphism of posets $\mathrm{Ch}^*(I^n) \cong \mathrm{Sal}(n)^{\mathrm{op}}$. Consequently $|\mathrm{Ch}^*(I^n)| \simeq \mathrm{Conf}(n, \mathbb{C})$ and $\pi_1 |\mathrm{Ch}^*(I^n)| \cong P_n$.*

The proof is straightforward: send $\mathrm{Run}(I^n)$ to topes, and observe that the composition operator agrees with the projection. This is machine-checked in [aut26].

Remark 5.12. $|\mathrm{Ch}(I^n)|$ is contractible, and that contractibility is the whole proof of Theorem 1.1. Theorem 5.11 says the complexification of the same object is a $K(P_n, 1)$. This is the sense in which the construction answers Theorem 1.1.

6 The braiding

You'll notice that the K side is not involved in the computation of the permutation, or the proofs of its functoriality. This comes from a rather nice fact about Grothendieck constructions.

Lemma 6.1. *The square*

$$\begin{tikzcd} \mathrm{Ch}^*(K) = \int(E(\rho) \circ W(K)^{\mathrm{op}}) \arrow[r, "Q"] \arrow[d, "\pi"'] \arrow[dr, phantom, "\lrcorner"] & \int E(\rho)|_{\square^\vee} = \square^\vee/\rho \arrow[d, "\pi"] \\ \mathrm{Ch}(K) \arrow[r, "W(K)"'] & \square^\vee \end{tikzcd}$$

is a strict pullback in **Cat**. *Here $\square^\vee/\rho$ is the comma category of serial wedges over ρ, formed along $\square^\vee \subseteq \square_* \to \square$; note ρ itself is not a serial wedge.*

Proof. By definition $\mathrm{Lines}(K)$ is the restriction of $E(\rho)|_{\square^\vee}$ along $W(K)$, and the category of elements of a restricted presheaf is the strict pullback of the category of elements along the restricting functor: both sides have as objects the pairs $(c \in \mathrm{Ch}(K),\ x \in E(\rho)(W(K)c))$, with the same morphisms. The identification $\int E(\rho)|_{\square^\vee} = \square^\vee/\rho$ is immediate: the category of elements of $\mathrm{Hom}_\square(-,\rho)$ is the comma category. $\square$

The map Q is just a projection to the first arrow of $\bigvee[1,\dots,1] \to \bigvee a \to K$. And ρ is the classifier for runs (Lemma 5.7). Since E is, in the contravariant case, the restricted Yoneda embedding, the right-hand corner is a comma category of wedges over ρ. Note that K does not appear on the right hand side at all.

Corollary 6.2. Ch^* *is a functor $\square_* \to$* **Cat**, *and the square of Lemma 6.1 identifies Q with $\mathrm{Ch}^*(!_K) : \mathrm{Ch}^*(K) \to \mathrm{Ch}^*(Z)$, where $!_K : K \to Z$ is the unique map to the final object.*

Proof. For $u : K \to L$ we have $W(L) \circ \mathrm{Ch}(u) = W(K)$, hence $\mathrm{Lines}(L) \circ \mathrm{Ch}(u)^{\mathrm{op}} = \mathrm{Lines}(K)$, which induces $\mathrm{Ch}^*(u)$ on categories of elements. Taking $L = Z$ and applying Lemma 6.4 gives the identification. $\square$

Lemma 6.3. *The functor Q is always faithful.*

Proof. Let $\bigvee[1,\dots,1] \to \bigvee a \to K$ and $\bigvee[1,\dots,1] \to \bigvee b \to K$ be two objects of $\mathrm{Ch}^*(K)$. Two morphisms f, g between these objects are equal exactly when they have the same underlying map $\bigvee a \to \bigvee b$, which happens if and only if $Q(f) = Q(g)$. So Q is indeed faithful. $\square$

But, we can do a nicer trick.

Lemma 6.4. *Let Z be the final precubical set. Then $\mathrm{Ch}^*(Z)$ is isomorphic to $\square^\vee/\rho$.*

Proof. We know Q is faithful already. To show Q is full, again we take two objects $\bigvee[1,\dots,1] \to \bigvee a \to Z$ and $\bigvee[1,\dots,1] \to \bigvee b \to Z$. Let f be a morphism in $\square^\vee/\rho$. That is, $f : \bigvee a \to \bigvee b$ forming a commuting triangle:

$$\begin{tikzcd} \bigvee a \arrow[r] \arrow[d, "f"'] & \rho \\ \bigvee b \arrow[ur] & \end{tikzcd}$$

But, since Z is the final precubical set, every map commutes in the triangle

$$\begin{tikzcd} \bigvee a \arrow[r] \arrow[d, "f"'] & Z \\ \bigvee b \arrow[ur] & \end{tikzcd}$$

So Q is full.

On objects, every wedge map $\bigvee a \to \rho$ is the same data as a $\bigvee[1,\dots,1] \to \bigvee a$ by Lemma 5.7, and every $\bigvee a$ has a unique map $\bigvee a \to Z$. So Q is a bijection on objects, and hence an isomorphism. $\square$

6.1 Braid words from permutations

But we need a device for building braid words from permutations, which brings us to the Garside germ presentation of braids [Gar69; Deh+15]. For $\sigma \in \Sigma_N$ write

$$\mathrm{Inv}(\sigma) := \{\, (i,j) \in [N] \times [N] \mid i < j \text{ and } \sigma(j) < \sigma(i) \,\}$$

for its set of *inversions*, and $\ell(\sigma) := |\,\mathrm{Inv}(\sigma)|$ for its *length*. We use the same notation for a bijection between two finite linearly ordered sets, the inversions being taken with respect to those orders; the only case needed is a coordinate map $f_* : \coprod_i [a_i] \to \coprod_j [b_j]$ with the lexicographic orders of Definition 3.3.

Definition 6.5. Let $\mathbf{Braid}^+$ be the category with objects the naturals and $\mathrm{Hom}(N,N) = B_N^+$ the positive braid monoid on N strands in its *germ presentation*: one generator $[\sigma]$ for each $\sigma \in \Sigma_N$, subject to

$$[\sigma][\sigma'] = [\sigma\sigma'] \qquad \text{whenever} \qquad \ell(\sigma\sigma') = \ell(\sigma) + \ell(\sigma').$$

Write **Braid** for the corresponding category of braid *groups*, $\mathrm{Hom}(N,N) = B_N$; the canonical functor $\mathbf{Braid}^+ \to \mathbf{Braid}$ is faithful by Garside's theorem [Gar69].

The key fact is a "no double crossings" feature of Ch^*: along a composite, no two events cross each other twice. Let's first state this in pure combinatorics.

Lemma 6.6. *Let f and g be permutations of $[N]$, and suppose that* $\mathrm{Inv}(f) \subseteq \mathrm{Inv}(g \circ f)$. *Then* $\ell(g \circ f) = \ell(g) + \ell(f)$.

Proof. Every element of $\mathrm{Inv}(g \circ f)$ either lies in $\mathrm{Inv}(f)$ or not, so

$$\mathrm{Inv}(g \circ f) \;=\; \big(\mathrm{Inv}(g \circ f) \cap \mathrm{Inv}(f)\big) \;\sqcup\; \big(\mathrm{Inv}(g \circ f) \setminus \mathrm{Inv}(f)\big).$$

By hypothesis $\mathrm{Inv}(f) \subseteq \mathrm{Inv}(g \circ f)$, so the first set equals $\mathrm{Inv}(f)$ and contributes $\ell(f)$.

For the second, a pair $(k,l) \notin \mathrm{Inv}(f)$ has $f(k) < f(l)$, and then $(k,l) \in \mathrm{Inv}(g \circ f)$ iff $g(f(k)) > g(f(l))$, i.e. iff $(f(k), f(l)) \in \mathrm{Inv}(g)$. Since f is a bijection, $(k,l) \mapsto (f(k), f(l))$ restricts to a bijection

$$\mathrm{Inv}(g \circ f) \setminus \mathrm{Inv}(f) \;\xrightarrow{\sim}\; \{(p,q) \in \mathrm{Inv}(g) \;:\; f^{-1}(p) < f^{-1}(q)\}.$$

This target is all of $\mathrm{Inv}(g)$: otherwise take $(p,q) \in \mathrm{Inv}(g)$ and set $k = f^{-1}(q) < l = f^{-1}(p)$; then $f(k) = q > p = f(l)$ gives $(k,l) \in \mathrm{Inv}(f)$, while $g(f(k)) = g(q) < g(p) = g(f(l))$ gives $(k,l) \notin \mathrm{Inv}(g \circ f)$, contradicting $\mathrm{Inv}(f) \subseteq \mathrm{Inv}(g \circ f)$. Hence the second set contributes $\ell(g)$, and $\ell(g \circ f) = \ell(f) + \ell(g)$. $\square$

The two halves of that feature are the following pair, which between them say that crossings happen only inside the fibres of blk, and that nothing crosses twice inside a fibre.

Lemma 6.7 (crossings are intra-block)**.** *Let $g : (\bigvee c, r) \to (\bigvee b, q)$ in $\mathrm{Ch}^*(K)$ and $(k, l) \in \mathrm{Inv}(\sigma(g))$. Then $\sigma(g)(k)$ and $\sigma(g)(l)$ lie in the same fibre of $\mathrm{blk} \circ q_*$.*

Proof. Put $u := \sigma(g)(k)$ and $v := \sigma(g)(l)$, so $k < l$ and $v < u$. Since $q_* \circ \sigma(g) = g'_* \circ r_*$ we have $q_*(u) = g'_*(r_*(k))$ and $q_*(v) = g'_*(r_*(l))$, so by Definition 3.4 and the monotonicity of $\mathrm{blk}(g')$ and of $\mathrm{blk} \circ r_* = \mathrm{blk}(r)$ (Lemma 3.6),

$$\mathrm{blk}(q_*(u)) = \mathrm{blk}(g')\big(\mathrm{blk}(r_*(k))\big) \;\le\; \mathrm{blk}(g')\big(\mathrm{blk}(r_*(l))\big) = \mathrm{blk}(q_*(v)),$$

using $k < l$. On the other hand $v < u$ and monotonicity of $\mathrm{blk} \circ q_* = \mathrm{blk}(q)$ (Lemma 3.6) give $\mathrm{blk}(q_*(v)) \le \mathrm{blk}(q_*(u))$. Hence $\mathrm{blk}(q_*(u)) = \mathrm{blk}(q_*(v))$. ☐

Theorem 6.8. *The map $f \mapsto [\sigma(f)]$ yields a functor $\mathrm{Conc} : \mathrm{Ch}^*(K) \to \mathbf{Braid}^+$, natural in K.*

Proof. An object goes to the braid object on its number of coordinates, well defined by Lemma 3.6, and $\sigma(\mathrm{id}) = \mathrm{id}$. For composition we need $[\sigma(f)][\sigma(g)] = [\sigma(f) \circ \sigma(g)]$, which by the germ presentation holds once $\ell(\sigma(f) \circ \sigma(g)) = \ell(\sigma(f)) + \ell(\sigma(g))$, and by Lemma 6.6 it is enough to prove $\mathrm{Inv}(\sigma(g)) \subseteq \mathrm{Inv}(\sigma(f) \circ \sigma(g))$ for composable $g : (\bigvee c, r) \to (\bigvee b, q)$ and $f : (\bigvee b, q) \to (\bigvee a, p)$.

Let $(k, l) \in \mathrm{Inv}(\sigma(g))$ and put $u := \sigma(g)(k)$, $v := \sigma(g)(l)$, so $v < u$. By Lemma 6.7 the positions u and v share a fibre of $\mathrm{blk} \circ q_*$, so Lemma 5.10 applied to f turns $v < u$ into $\sigma(f)(v) < \sigma(f)(u)$, which is exactly $(k, l) \in \mathrm{Inv}(\sigma(f) \circ \sigma(g))$.

For naturality in K, note that σ reads only the underlying wedge data, so $\mathrm{Conc}_K = \mathrm{Conc}_Z \circ \mathrm{Ch}^*(!_K)$ by Corollary 6.2. ☐

This looks a bit silly, because all we've done is proven our braid words have no cancellation. But that only means we found a nice basis, and in fact it means we're working in the positive braid monoid, where by Garside's theorem [Gar69] the map $B_N^+ \hookrightarrow B_N$ loses no information. We recover the full braiding structure by lifting to the free groupoid.

Definition 6.9. Let $\mathrm{Free}(\mathrm{Ch}^*(K))$ be the free groupoid on $\mathrm{Ch}^*(K)$ — its localisation at all morphisms — and let

$$\mathrm{Conc}(K) : \mathrm{Free}(\mathrm{Ch}^*(K)) \longrightarrow \mathbf{Braid}$$

be the unique extension of Theorem 6.8 along $\mathrm{Ch}^*(K) \to \mathrm{Free}(\mathrm{Ch}^*(K))$, which exists because **Braid** is a groupoid. We call $\mathrm{Conc}(K)$ — or, once a component is fixed, the induced homomorphism on vertex groups — the *concurrency braid* of K.

Proposition 6.10. *For any precubical set K, if there is an $f : I^2 \to K$, then $\mathrm{Conc}(K)$ is nontrivial.*

Proof. It suffices to find a morphism in $\mathrm{Ch}^*(K)$ whose permutation is not the identity. Consider the map $p : \bigvee[1,1] \to \bigvee[2]$ going along the bottom edge and then the right edge, and the map $q : \bigvee[1,1] \to \bigvee[2]$ going along the left edge and then the top edge. We have

$$\begin{tikzcd}
\bigvee[1,1] \arrow[r, "\mathrm{id}"] \arrow[d, dashed, "\sigma(p)"'] & \bigvee[1,1] \arrow[d, "p"] \arrow[dr] & \\
\bigvee[1,1] \arrow[r, "q"'] & \bigvee[2] \arrow[r, "f"'] & K
\end{tikzcd}$$

This is a morphism of $\mathrm{Ch}^*(K)$ from $(\bigvee[1,1] \xrightarrow{f\circ p} K,\ \mathrm{id})$ to $(\bigvee[2] \xrightarrow{f} K,\ q)$: the condition $x = \mathrm{Lines}(p)(y)$ is automatic because $\mathrm{Run}(\bigvee[1,1])$ is a singleton. Its permutation is $\sigma(p) = q_*^{-1} \circ p_* \circ \mathrm{id} = \{1 \mapsto 2,\ 2 \mapsto 1\}$, because p traverses direction 1 then direction 2 while q traverses direction 2 then direction 1. This has an inversion, so $[\sigma(p)]$ is not the identity braid. $\square$

Remark 6.11 (Why there is no coherent choice of run). $\pi : \mathrm{Ch}^*(K) \to \mathrm{Ch}(K)$ is a discrete fibration, and for a fixed K it may well admit sections. What it does not admit is a family of sections $s_K : \mathrm{Ch}(K) \to \mathrm{Ch}^*(K)$ natural in K — and it had better not. Suppose it did. Applying naturality to the unique map $K \to Z$ and using Corollary 6.2, $\mathrm{Conc} \circ s_K$ would equal $G \circ W(K)$ for the single functor $G := \mathrm{Conc} \circ s_Z$ on $\square^\vee \cong \mathrm{Ch}(Z)$, so the braid data would be a function of the chain alone. For $K \subseteq I^n$ that factors through $\mathrm{Ch}(I^n)$, whose nerve is contractible, and the invariant would die exactly as in Theorem 1.1. So the content of Ch^* is precisely that the run *cannot* be chosen coherently: that incoherence is the braiding. Turning this into a theorem — fixing the class of K over which naturality is demanded, and the sense in which the resulting invariant is trivial — is Section 7(3).

Remark 6.12 (Relation to IB_n). Two differences from the invariants of [PZ21, §6] are worth naming. Theirs is a conjugacy class of homomorphism out of the fundamental group of an execution space, so it needs a basepoint and is only defined up to conjugacy; Conc is a functor on a category, so it needs neither. And theirs lands in B_n, the fundamental group of the *unordered* configuration space, whereas Theorem 5.11 places ours over the ordered one. The Σ_n-quotient relating the two is Section 7(2).

7 Next steps

1. **What does $|\mathrm{Ch}^*(K)|$ compute?** What gives Ch its meaning is [Zie20, Thm. 7.6]: its nerve is the execution space. We have no such statement for Ch^* beyond $K = I^n$, where Theorem 5.11 identifies it with a configuration space. Is there a space of "complexified d-paths" on $|K|$ whose homotopy type is $|\mathrm{Ch}^*(K)|$, naturally in K? This is the main open question, and much of what follows is downstream of it.
2. **Recovering B_n.** Σ_n acts on $\mathrm{Sal}(n)$ with quotient the Salvetti complex of the unordered arrangement, and the corresponding covering realises $P_n \leq B_n$. Carrying this through on the Ch^* side should recover the invariants of [PZ21, §6] as a quotient of ours, making Remark 6.12 precise. The covering and the deck sequence are formalised in [aut26].
3. **The obstruction.** Make Remark 6.11 a theorem: there is no section of $\pi : \mathrm{Ch}^*(-) \to \mathrm{Ch}(-)$ natural in K. The sketch given there reduces it to the triviality of any Ch-level braid invariant on sculptures, which is Theorem 1.1; what needs care is the quantifier over K and the notion of triviality.
4. **Subdivision and Markov moves.** The strand count of Conc is $N = \mathrm{alt}(\mathrm{fin}_K)$, the number of events along an execution, so it changes when K is subdivided. This is the same indexing used in [PZ21, §6], where the invariant is defined one execution at a time, so it is a feature of the setting rather than of this construction; but it does mean Conc is an invariant of the cubulation, not of the directed homotopy type. Looking at braid closures instead, we can relate braids with different numbers of strands via the Markov moves. I believe there is a connection to subdividing K, but more work is required to formalize this.
5. **A chain complex.** The category $\mathrm{Ch}^*(K)$ is graded by the shape of the wedges: (the sum of the dimensions) $-$ (the length of the list). Each morphism should decompose into a sequence

of degree 1 morphisms. I think this is the face relation of a CW complex, but I'm not certain, and I also think there is a sensible boundary map which gives us a chain complex. Unlike the rest of the paper, none of this is formalised.

6. **Examples.** Beyond I^n and Z nothing is computed. A PV-program example — the Swiss flag, say — in which Conc distinguishes executions would do a lot to substantiate the claim that this is an invariant of concurrent programs.

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